Github link to main branch: https://github.com/uob-team-project/team23

Entity Name: **GlobalTopsongs**

Purpose: The GlobalTopsongs entity stores and manages global top songs data, allowing users to view monthly/yearly popular songs based on metrics such as play count, release year, and artist information. It also provides functionality for filtering and sorting songs, enabling users to explore trends in music popularity over time, and easily navigate through different views of the top songs.

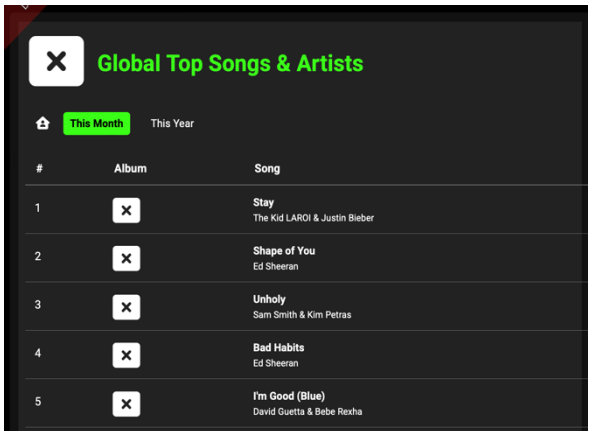
Key Attributes:

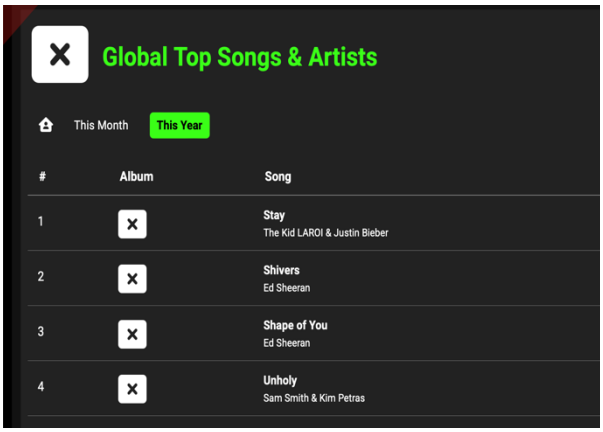
- id: Unique identifier
- title: Song title
- artist: Artist name
- playCount: Number of times the song has been played
- monthYear: Date of monthly or yearly song popularity data collection

User Features:

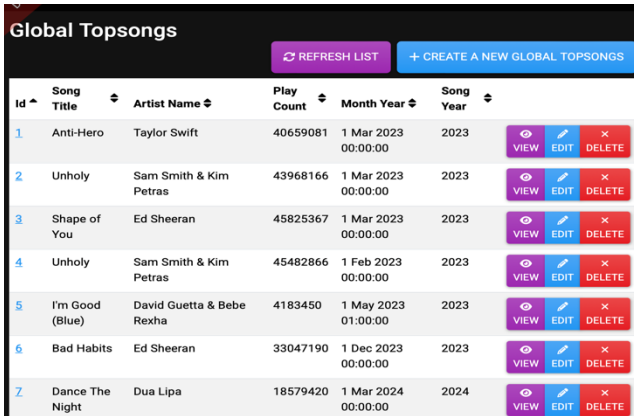
1. View Monthly/Yearly Top Songs
    - Users can browse a list of globally popular songs filtered by month and year.
    - The system provides sorting and pagination to enhance usability.
2. Create New Songs Entries
    - Only admins can add a new song entry to the database
    - The system ensures that all required fields (title, artist, play count, and date) are correctly filled before submission.
3. Read (View) Song Details
    - Clicking on id in the list takes the user to a detailed view, displaying all its attributes.
    - Users can navigate back to the main list from the detailed view.
4. Update (Edit) Existing Songs
    - Only admins can edit the details of an existing song, updating information such as title, artist name, or play count.
    - The system validates input fields to prevent incorrect data entry.
5. Delete Songs
    - Only admin can delete a song entry from the system.
    - Regular users cannot delete songs to ensure data integrity.
6. Automated Data Updates
    - The system updates the list of top songs automatically every month and year.
    - Users do not need to manually refresh or upload data for periodic updates.

Frontend Components and Description

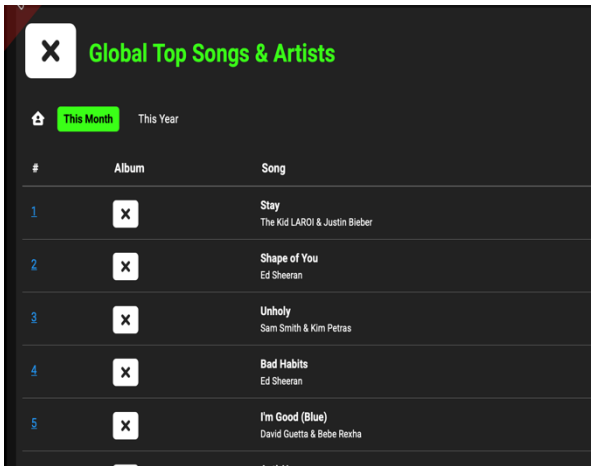


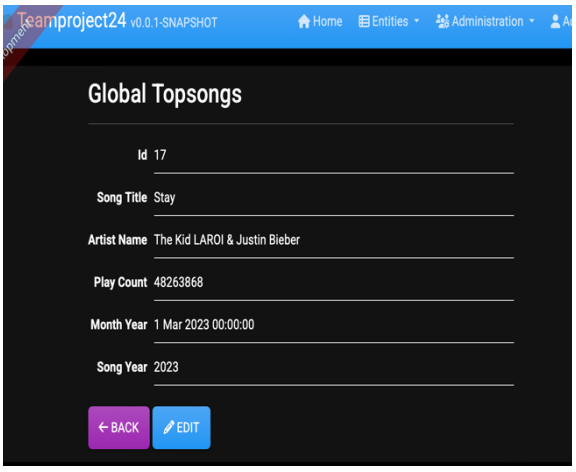


TopSongsComponent – Offers filtering functionality so users can view top songs by month or year. Helps in organizing data for better analysis.

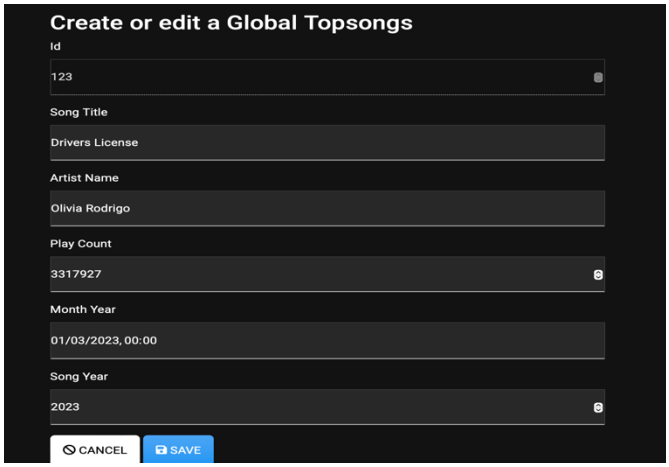


GlobalTopsongsComponent – Displays the list of top songs. Sorting, editing, deleting is available from this view. Includes pagination for better navigation.





GlobalTopsongsDetailComponent - When clicking the id of the song it moves to a page providing a detailed view of a selected song, showing all its attributes. Users can return to the list from this screen.



GlobalTopsongsUpdateComponent – Contains a form that allows admin to create a new song entry or modify an existing one. Input validation ensures data accuracy.

GlobalTopsongsDeleteDialogComponent – Displays a confirmation dialog when a user (admin) attempts to delete a song. Prevents accidental deletions.

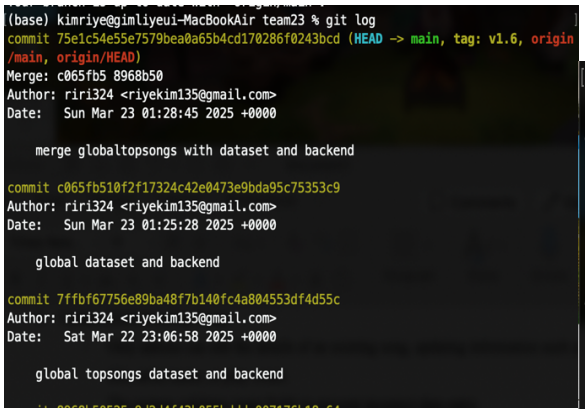
Backend Constraints

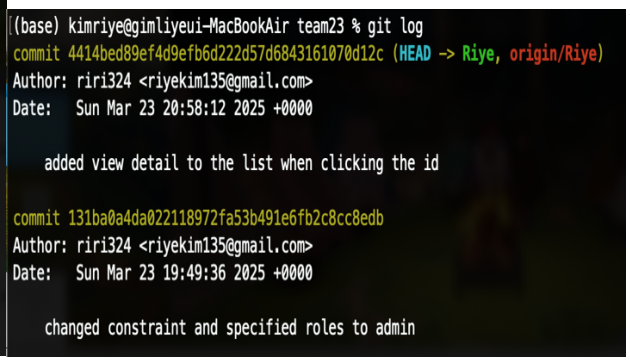
1. Data Integrity Rules
    - The playCount value must always be greater than or equal to 0 to prevent invalid song statistics.
    - The monthYear field must fall within the valid range of 2000 to 2100, ensuring historical and future data accuracy.

2. Access Control for Deletion
    - Only admin have the ability to delete a song entry.
    - Regular users cannot delete songs, protiecting against accidental or unauthorized data removal.

3. Automated Data Updates
    - The system automatically updates the top song list monthly and yearly, so users always see the most relevant data.
    - There is no need for manual intervention to refresh or add periodic data updates.

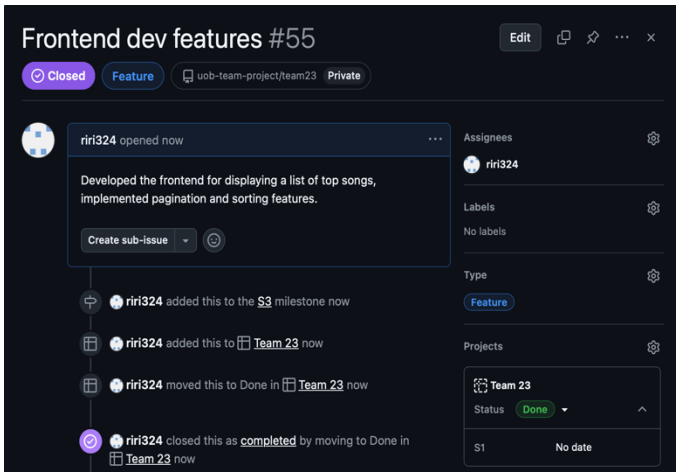
GitHub Commit List





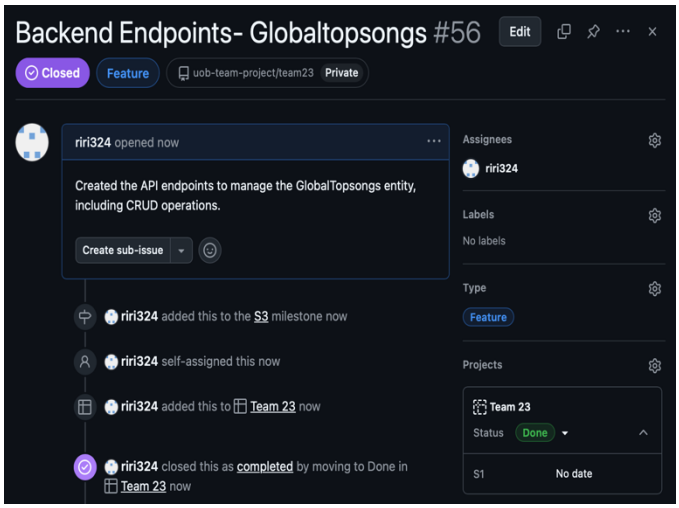
Date	Commit Message	Description	Affected Files
2025-03-23	“changed constraint and specified roles to admin”	Improved the constraints and changed (improved) edit, delete, update functions to the roles from users to admin	- Global-topsongs-constraints.txt - GlobalTopsongsResource.java
2025-03-23 Riye(branch)	“added view detail to the list when clicking the id”	When clicking the id it directs to the page to view song details	- app.routes.ts - global-topsongs-detail.component.html - global-topsongs-detail.component.spec.ts - global-topsongs-detail.component.ts - global-topsongs.routes.ts - top-song.component.html - top-song.component.ts
2025-03-23 Riye(branch)	“global dataset and backend”	Connected the .csv with the frontend and backend	- app.routes.ts
2025-03-22 main	“global topsongs dataset and backend”	Connected the .csv with the frontend and backend	- GlobalTopsongs.java - GlobalTopsongsDTO.java - 20250228154741_added_entity_GlobalTopsongs.xml - global_topsongs.csv - GlobalTopsongsRepository.java - global-topsongs.service.ts - global-topsongs.component.html - global-topsongs-detail.component.html - global-topsongs-update.component.html - global-topsongs.model.ts
2025-03-23 main	“global dataset and backend”	Connected the .csv with the frontend and backend	- app.routes.ts

Kanban Cards



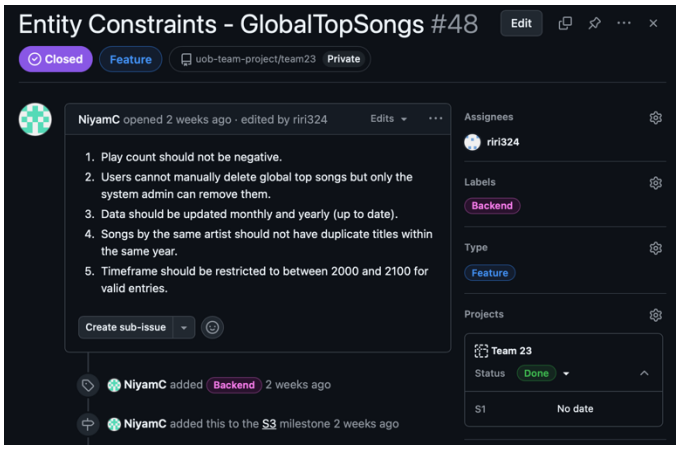
Frontend dev features

Developed the frontend for displaying a list of top songs, implemented pagination and sorting features.



Backend Endpoints

Created the API endpoints to manage the GlobalTopsongs entity, including CRUD operations.



Entity Constrains

Updated entity constraints.

Github link to main branch: https://github.com/uob-team-project/team23

Github link to Riye branch: https://github.com/uob-team-project/team23/tree/Riye

M3 App Demo 60s Script (GlobalTopsongs Entity)

I’m going to walk through the GlobalTopsongs entity in our app. This feature allows users to explore globally trending songs by displaying monthly and yearly top tracks based on play counts.

On the main GlobalTopsongs page, users can browse a list of trending songs. Each entry includes the id, album picture, song title and artist name. Clicking on a song id takes users to a detailed view, where they can see more information.

From an admin perspective, authorized users have additional functionality. Admins can edit song details, update metadata, or remove entries. These actions are controlled by role-based access to ensure only permitted users can modify the data.

The backend provides structured API endpoints for retrieving and managing the song data, enforcing validation constraints like required fields for song title and artist name. Additionally, Liquibase ensures the database schema remains consistent.

Below is the demo showcasing these features in action

Audio recording link: M3 app demo.m4a

https://bham-my.sharepoint.com/personal/rxk355_student_bham_ac_uk/Documents/M3%20app%20demo.m4a?csf=1&web=1&nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=FNEwl7

Slides Overview

Purpose:

- Tracks and displays globally trending songs (monthly/yearly)

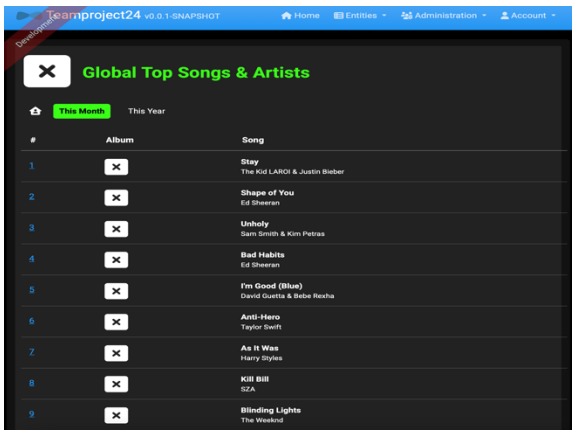
- Stores id, song title, artist, play count, released year, etc.

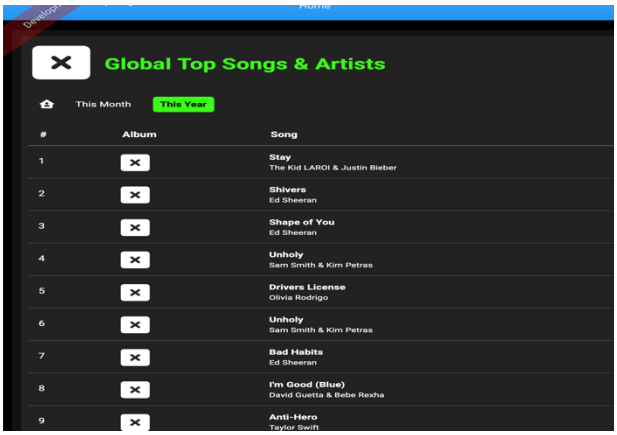
Key features:

- View trending songs.

- Admin only actions: Edit or delete songs

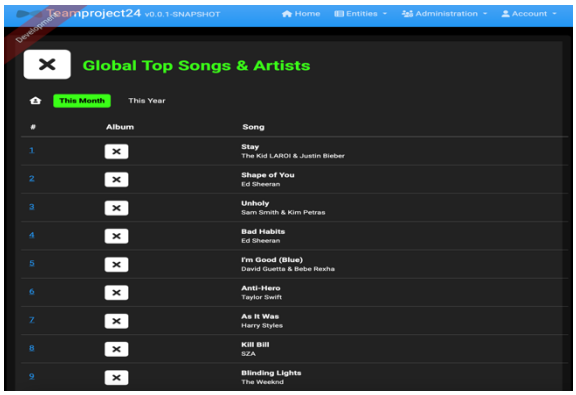
- Backend constraints: Role-based access control, required fields.



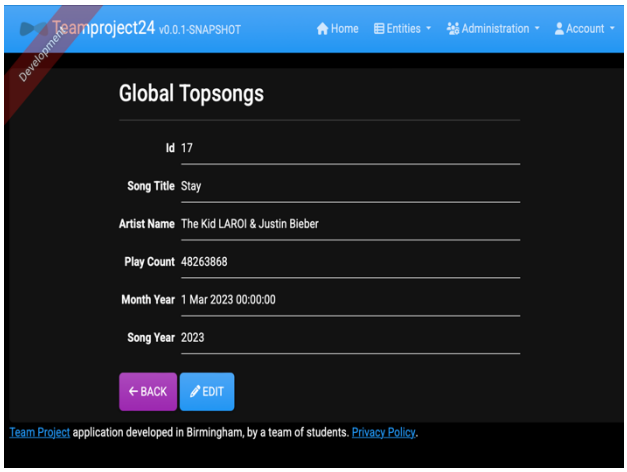


Demo Workflow

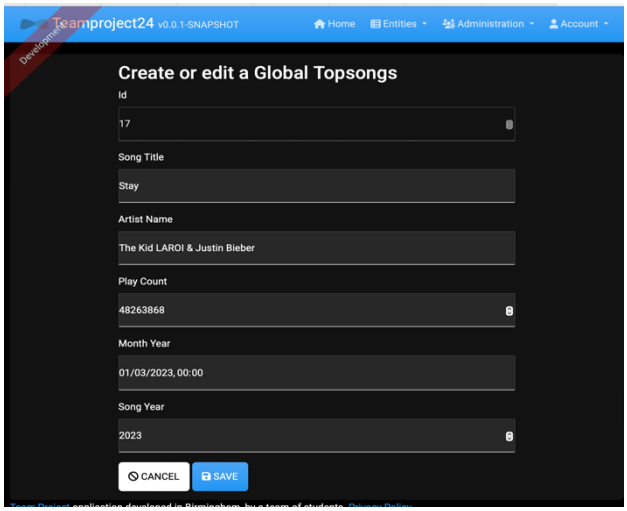
1. Navigate to GlobalTopsongs list (UI view of trending songs).



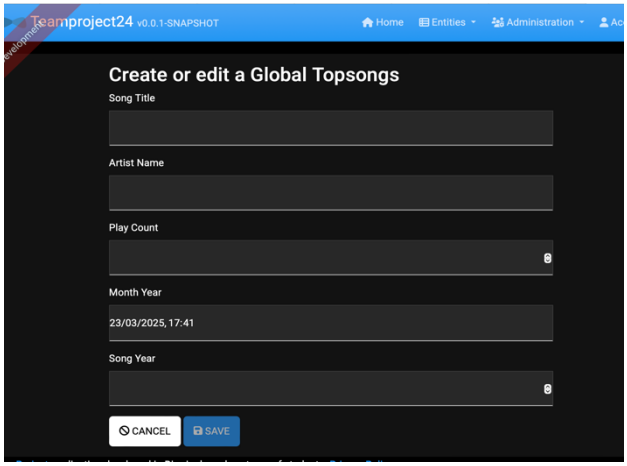
2. Click on a song to see details page.



3. Edit song form (Admin view) => Demonstrating role-based access



4. Backend API request => Example of a request to update a song.



M3 Accessibility (A11y) Demo – 60s Script

I’m going to walk through the accessibility features of the GlobalTopsongs entity in our app.

One of the key accessibility improvements we made is ensuring sufficient contrast between text and background colors. This helps users with visual impairments read content more easily. We followed WCAG guidelines to maintain a clear distinction between text and background elements, making the app more inclusive.

We also ensured that buttons and interactive elements have clear labels, so users can easily understand their functions. Additionally, we included Accessible Rich Internet Applications (ARIA) labels where necessary, allowing screen readers to interpret and announce elements correctly.

While the app doesn’t currently support full keyboard navigation, users can still navigate most content with standard browser tools. These accessibility enhancements help ensure a better experience for all users.

Below is the demo showcasing these features in action.

Audio recording link: M3 Accessibility (A11y) Demo.m4a

https://bham-my.sharepoint.com/personal/rxk355_student_bham_ac_uk/Documents/M3%20Accessibility%20(A11y)%20Demo.m4a?csf=1&web=1&nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAiOiJPbmVEcml2ZUZvckJ1c2luZXNzIiwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=MMncfe

Accessibility Feartures in GlobalTopsongs

Accessibility Improvements:

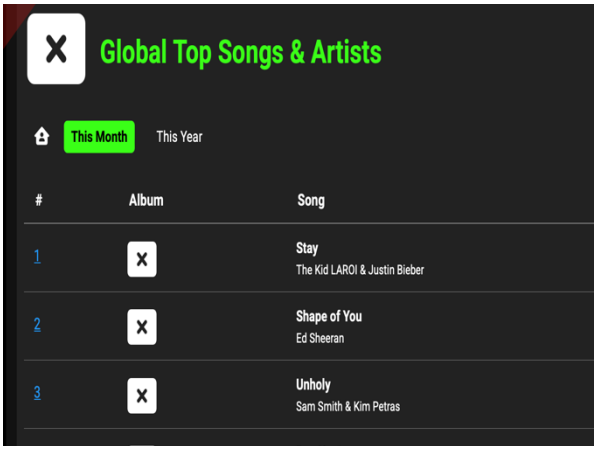
- Text to background contrast improved for better readability (WCAG compliant)

- ARIA labels added for enhanced screen reader support

- Clear labels on buttons and key interactive elements.

Demo Walkthrough:

1. Navigate to the GlobalTopsongs page (Highlighting the contrast improvements in text)



“Global Top Songs & Artist “ heading
- color: #39ff14;

Text and icon

- Color : white

“This Month” & “This Year” is clicked

- Text-color: Black, Button-color: #39ff14;

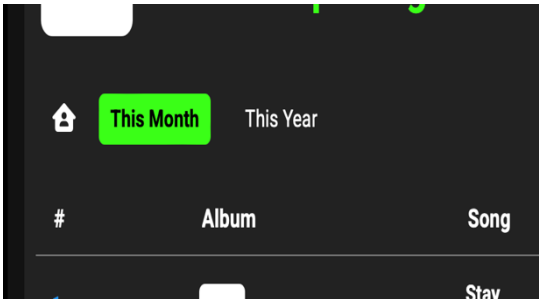
Background

- Color: Black

2. Click on a button => how ARIA labels support screen readers.

```
<div class="top-controls">
  <fa-icon [icon]="'house-user'" class="icon-button" (click)="goHome()" aria-label="Go to Home Page"></fa-icon>
  <span class="custom-tab-1" [class.active]="currentView === 'month'" (click)="showChart('month')"> This Month
  <span class="custom-tab" [class.active]="currentView === 'year'" (click)="showChart('year')"> This Year </spa
</div>
```

3. Show UI elements with clear labels



Time sheet

Team sheet Number/ID: Team23

Team member name: Reeyea Kim

Team representative (secratary) Member

Team meeting sign off date: 03.21.2025

Date from: 03/21/2025

Date until: 03/24/2025

Task	Date	Start time	End time	Total Hours
Entity Implementation(frontend and backend)	Mar-23	2:30 PM	9:55 PM	7:25
demo-prep: 60s script & 1-2 slides	Mar-24	12:00 PM	4:02 PM	4:02
				0:00
				0:00
				0:00
				0:00
				0:00

Total Hours **11:27**